# Max-Flow-Min-Cut-Theorem

**Theorem.** The value of a maximum $s$–$t$-flow and the weight of a minimum $s$–$t$-cut are the same.

**Proof.** Special case of LP-Duality ...

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# Dual LP – Interpretation as ILP

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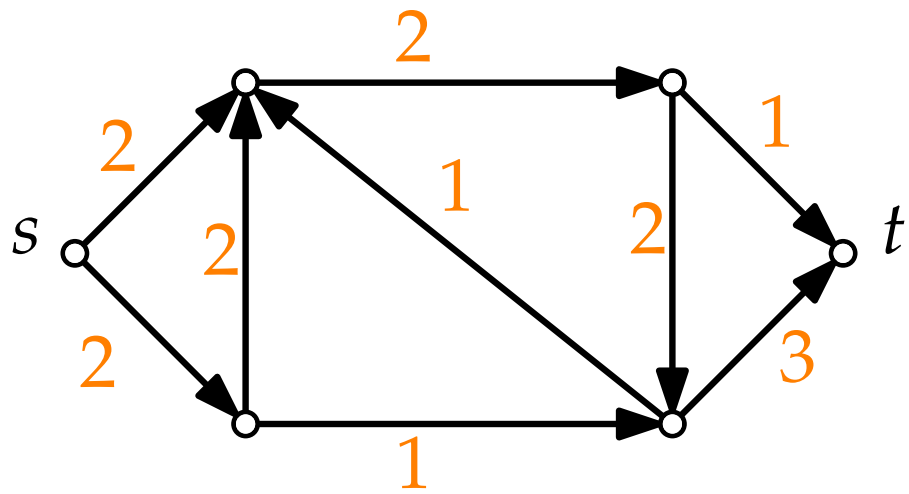
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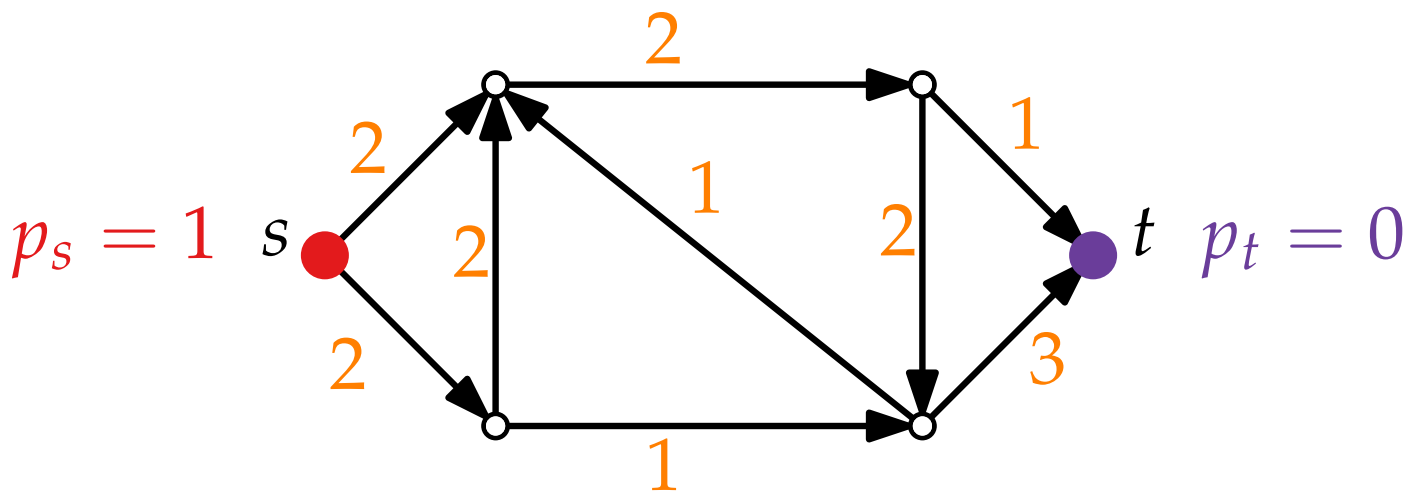
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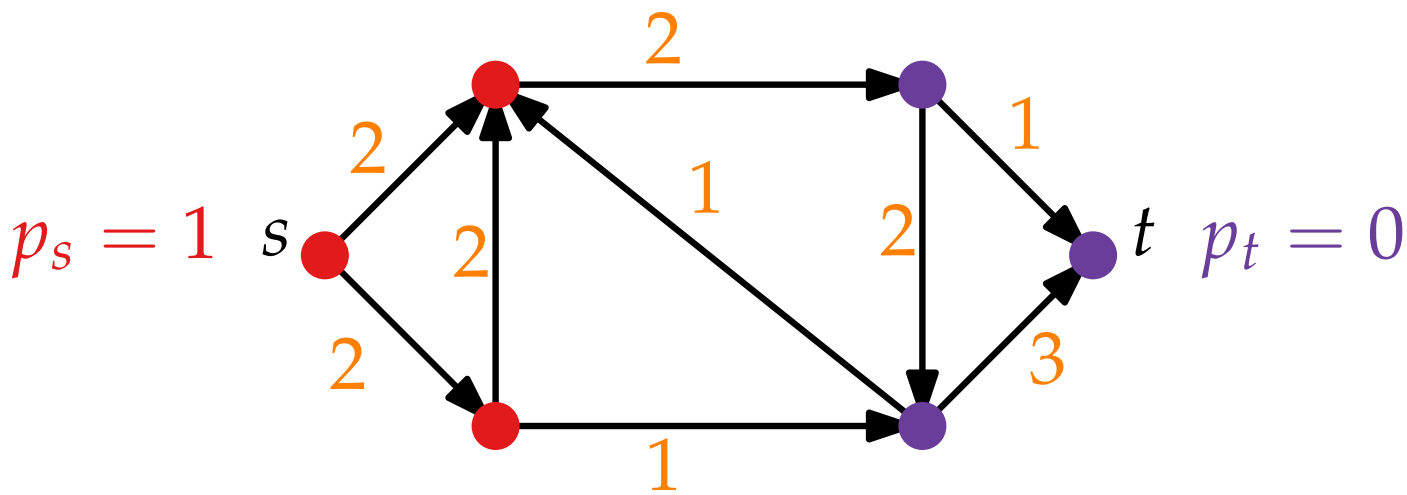
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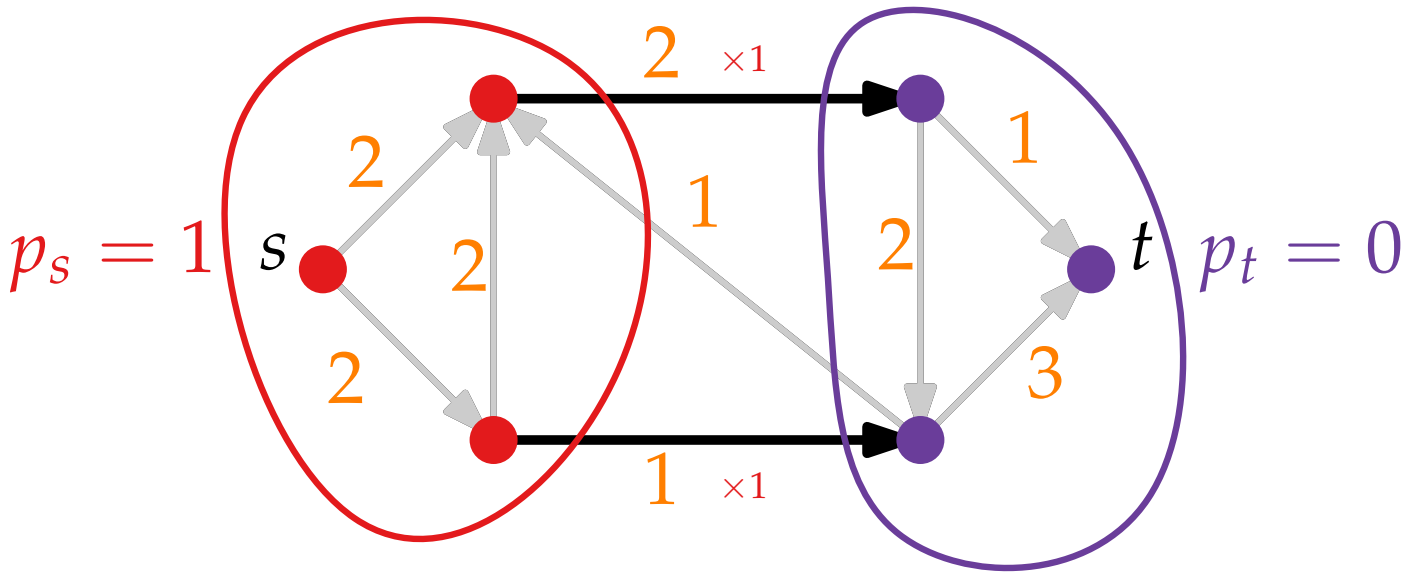
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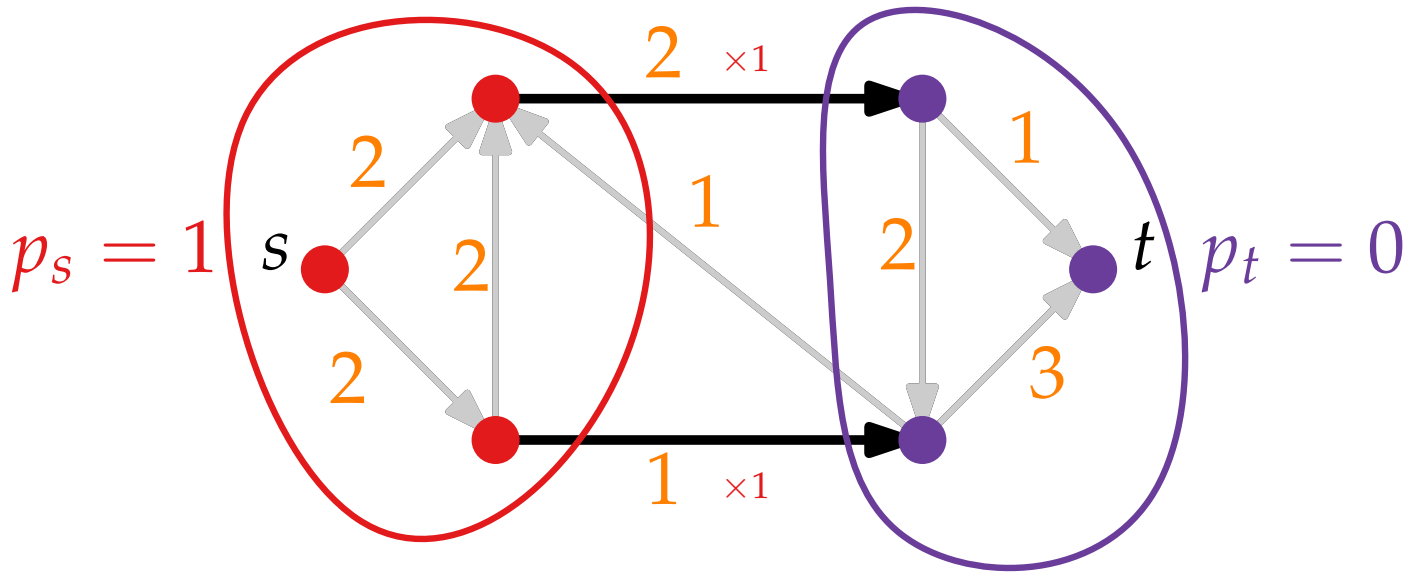
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equivalent to Min-Cut!

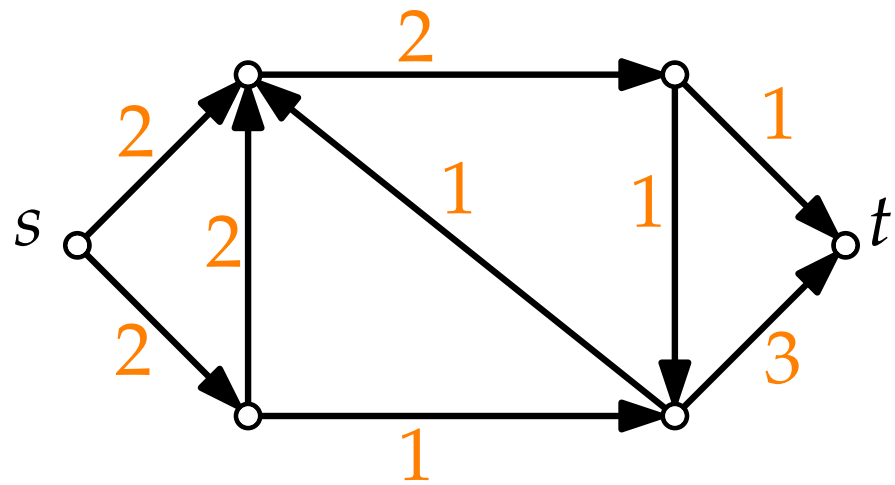


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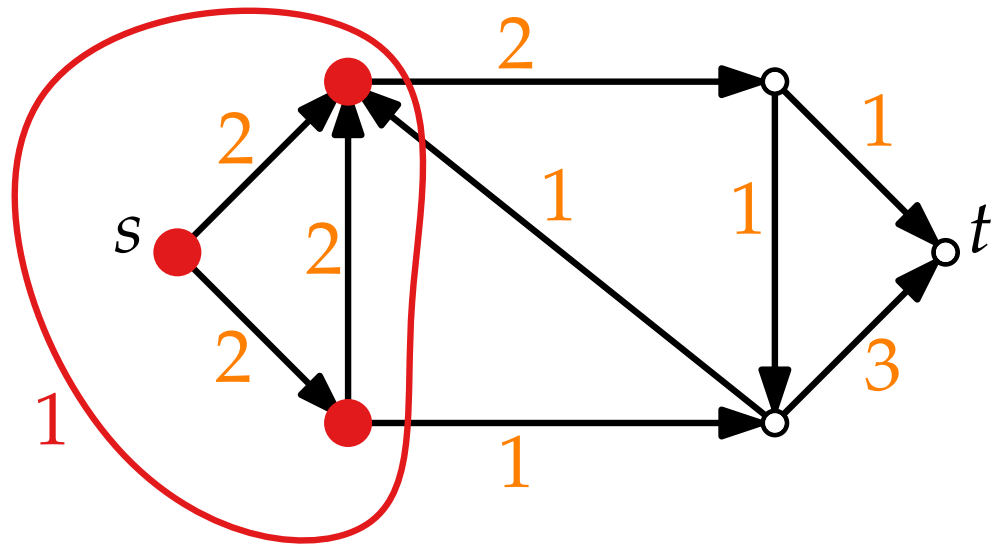
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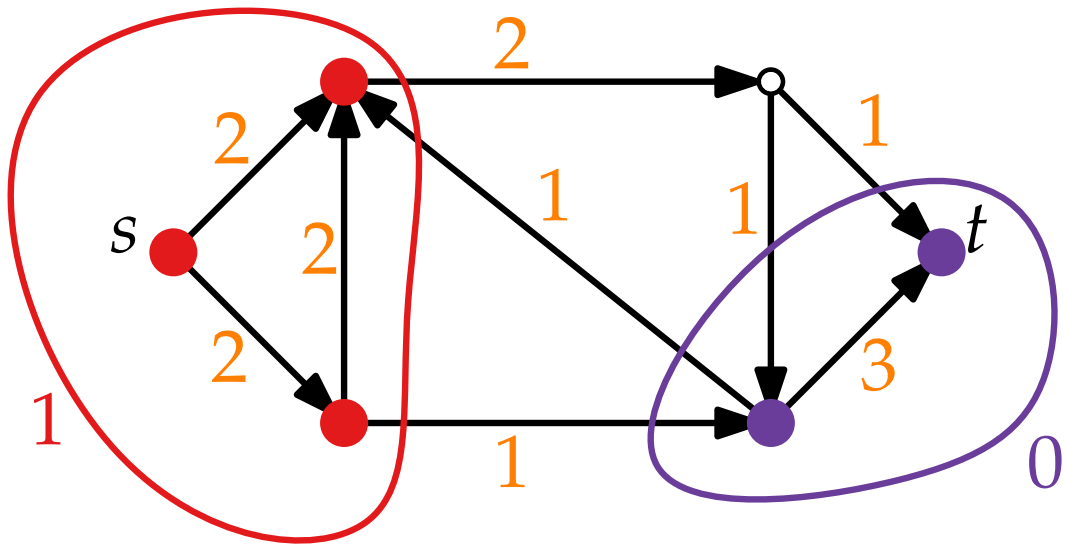
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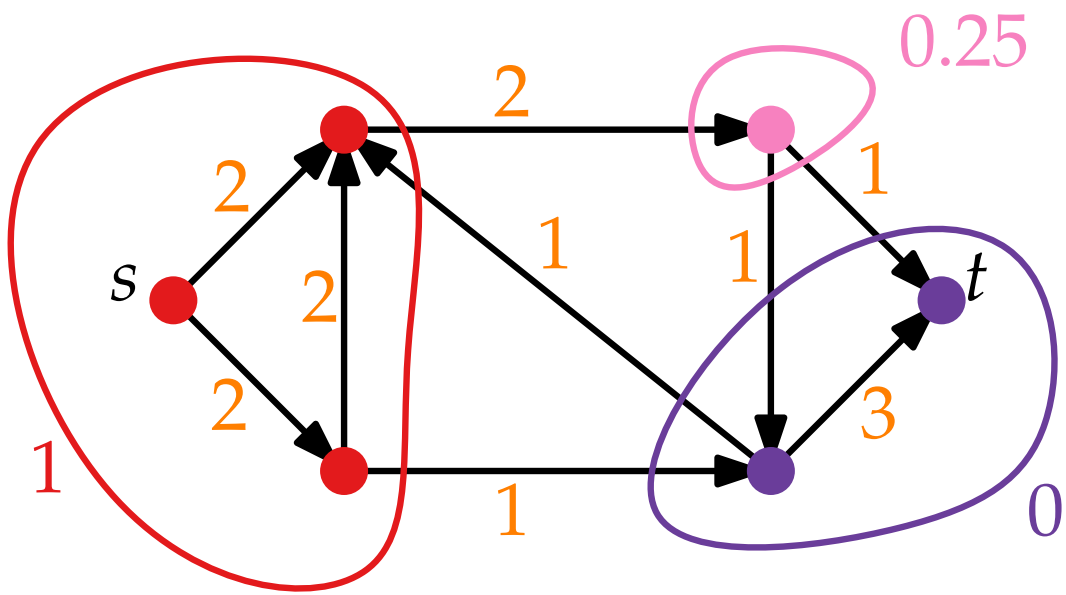
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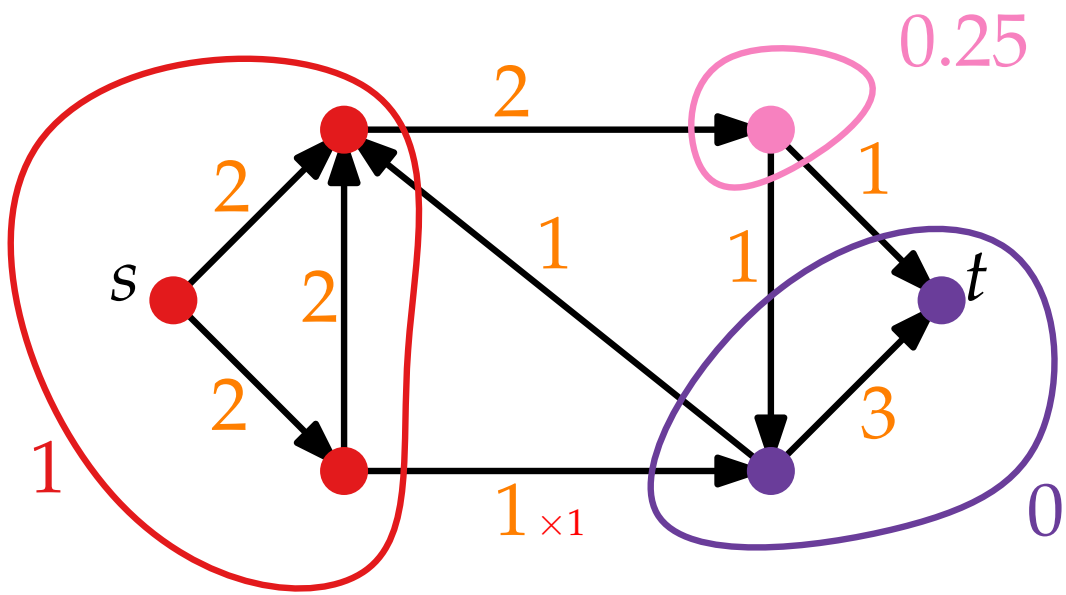
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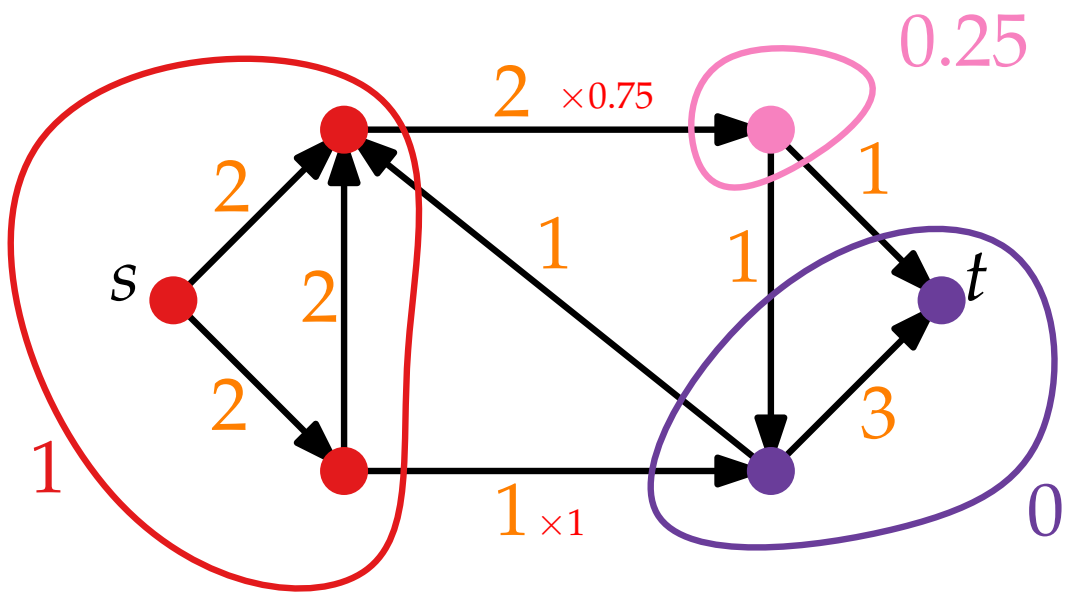
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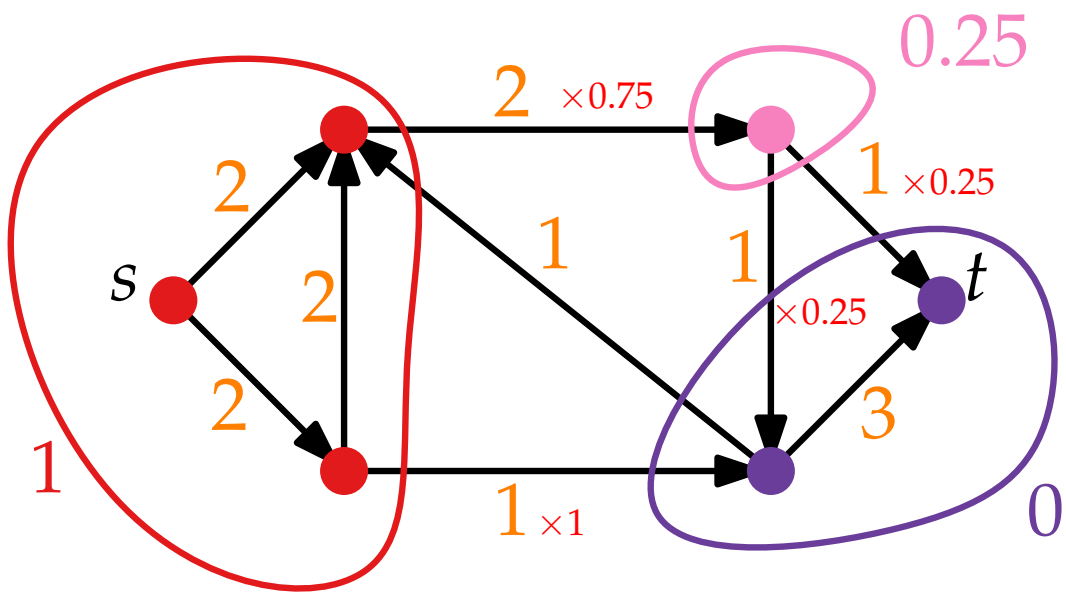
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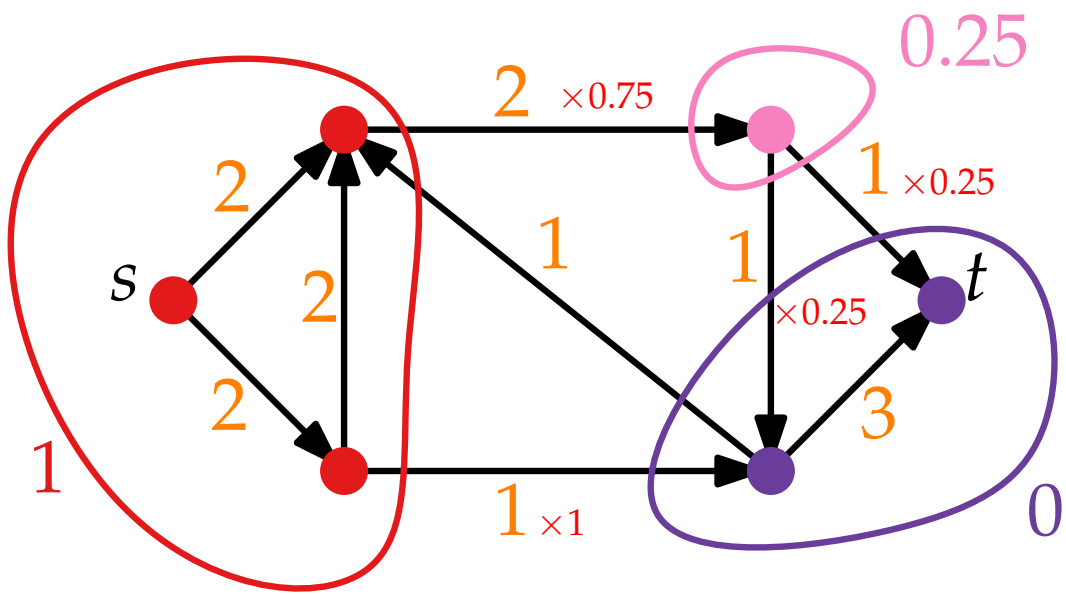
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Each $s$–$t$-path $s = v_0, \ldots, v_k = t$ has length $\geq 1$ w.r.t. $d$:

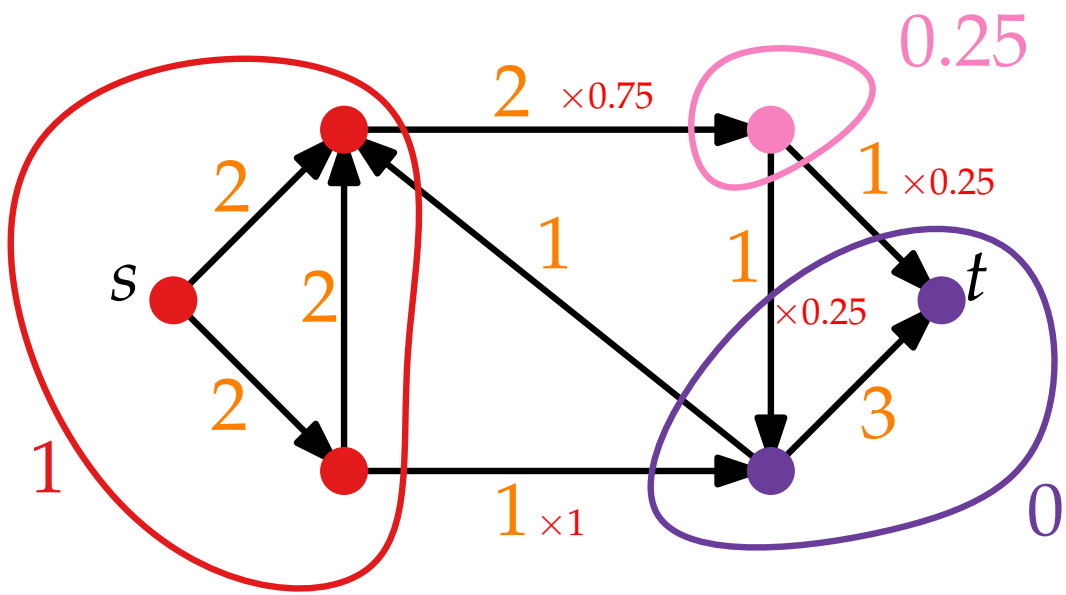


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| **subject to** | $d_{uv} - p_u + p_v \geq 0$ | $\forall (u,v) \in E \setminus \{(t,s)\}$ |
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| | $d_{uv} \geq 0$ | $\forall (u,v) \in E$ |
| | $p_u \geq 0$ | $\forall u \in V$ |

Each $s$–$t$-path $s = v_0, \ldots, v_k = t$ has length $\geq 1$ w.r.t. $d$:

$$\sum_{i=0}^{k-1} d_{i,i+1} \geq \sum_{i=0}^{k-1} (p_i - p_{i+1}) = p_s - p_t$$



# Dual LP – Complementary Slackness

**maximize** $f_{ts}$

**subject to**

$$f_{uv} \leq c_{uv} \qquad \forall (u,v) \in E \setminus \{(t,s)\}$$

$$\sum_{u:\,(u,v)\in E} f_{uv} - \sum_{z:\,(v,z)\in E} f_{vz} \leq 0 \qquad \forall v \in V$$

$$f_{uv} \geq 0 \qquad \forall (u,v) \in E$$

**minimize** $\sum_{(u,v)\in E\setminus\{(t,s)\}} c_{uv} \cdot d_{uv}$

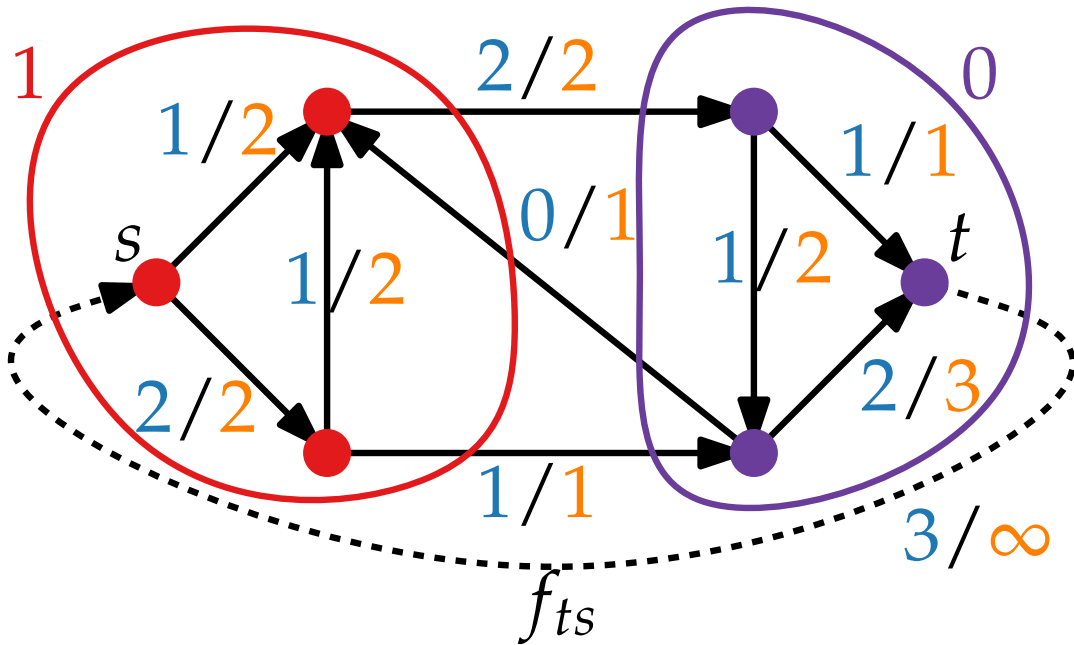
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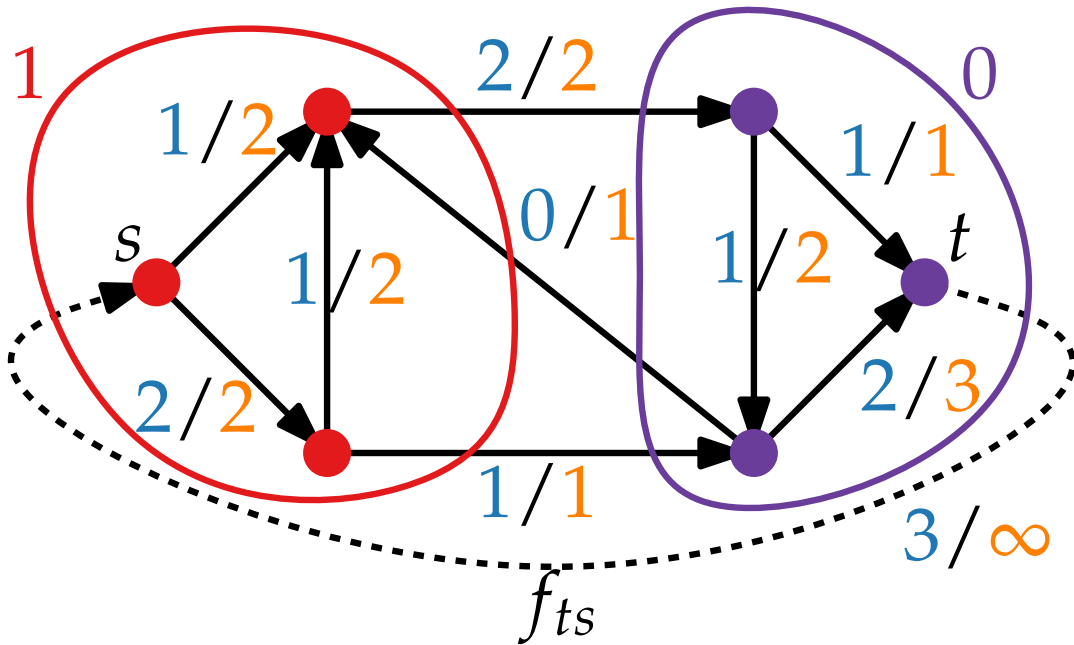
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For a max flow and min cut:



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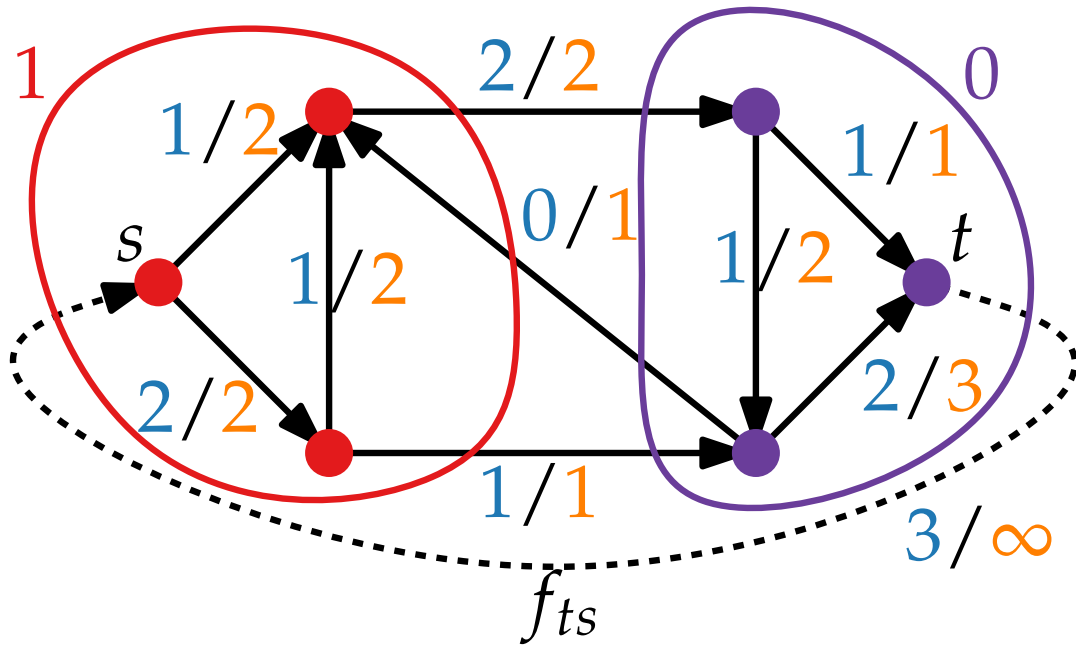
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For a max flow and min cut:

- For each forward edge $(u,v)$ of the cut: $f_{uv} = c_{uv}$.



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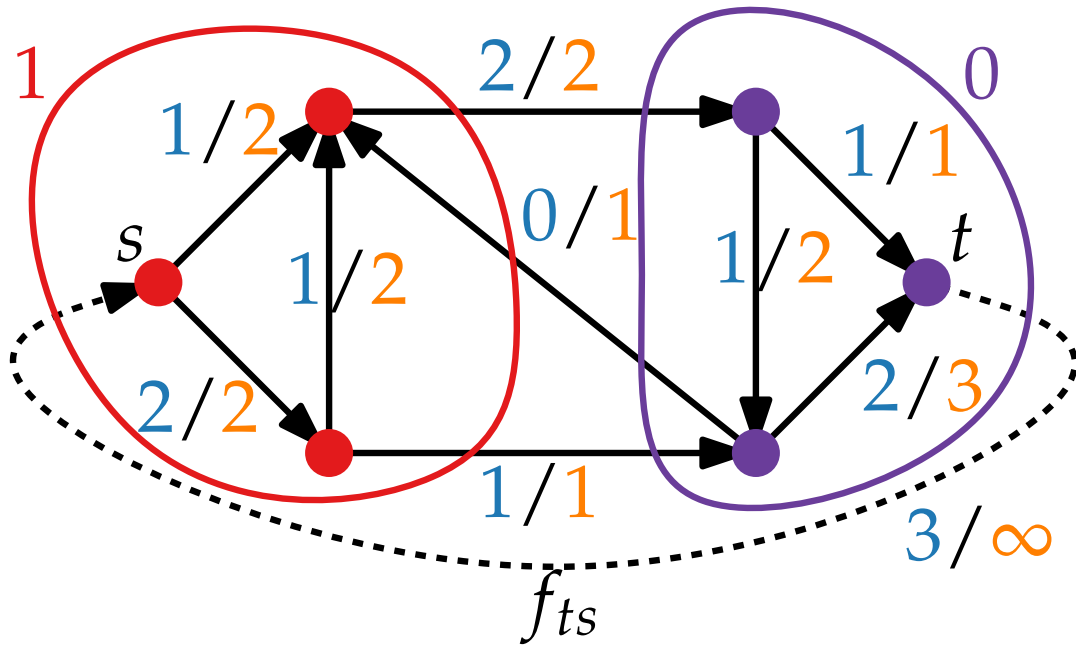
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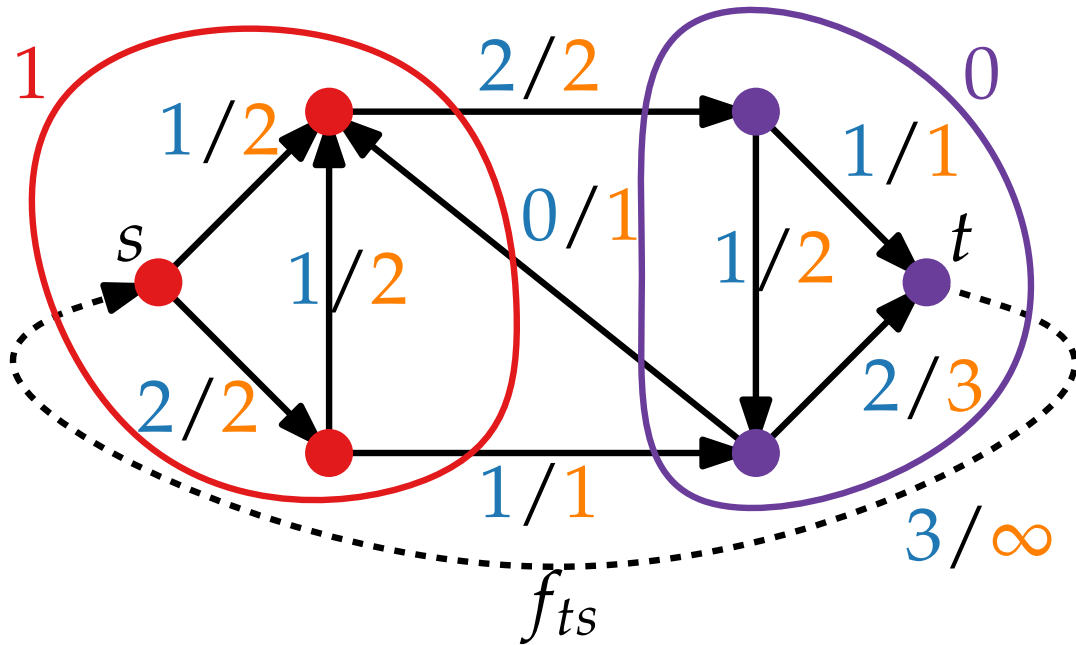
$$p_u \geq 0$$

**Primal CS**:
$\forall j$: Either $x_j = 0$ or $\sum_{i=1}^{m} a_{ij} y_i = c_j$

**Dual CS**:
$\forall i$: Either $y_i = 0$ or $\sum_{j=1}^{n} a_{ij} x_j = b_i$

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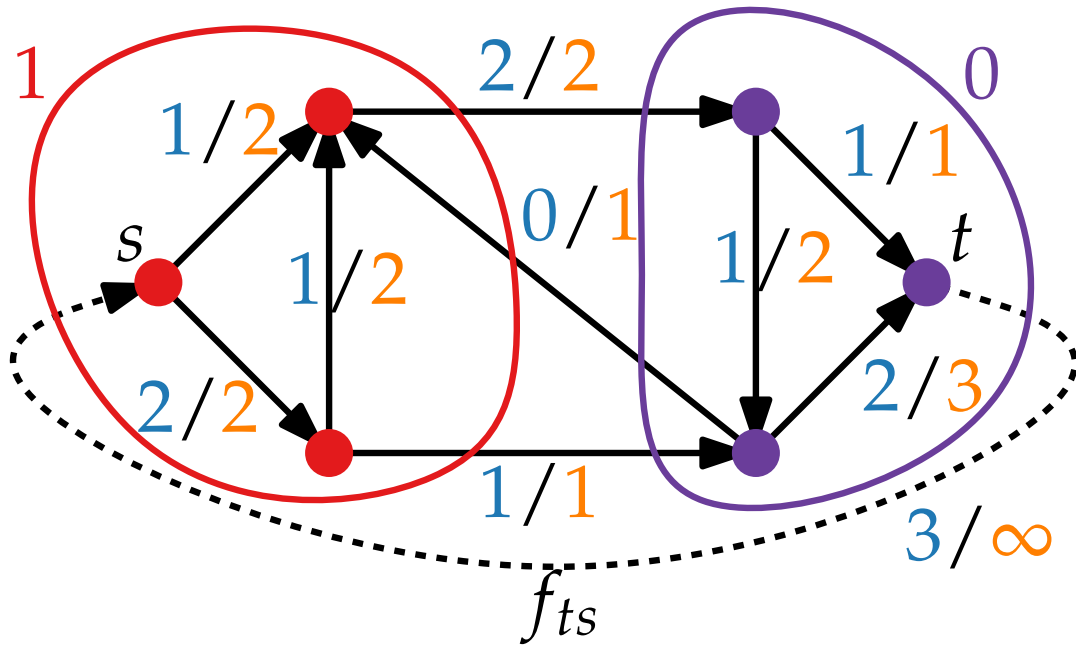
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For a max flow and min cut:

- For each forward edge $(u,v)$ of the cut: $f_{uv} = c_{uv}$.
  ($d_{uv} = 1$, so by dual CS: $f_{uv} = c_{uv}$.)
- For each backward edge $(u,v)$ of the cut: $f_{uv} = 0$.
  (Otherwise, by primal CS: $d_{uv} - 0 + 1 = 0$.)

